

Deep Learning - Methodology

	FR context	Grammar Rules	Number of Grammar Rules
FR1	Dataset Definition and Loading The grammar should support: Declarative specification of dataset sources , input data formats , and data loading configurations including file paths, data types , and input dimensions without requiring knowledge of underlying data handling mechanisms.	DatasetSection, DataSource(3 variants rules), DataFormat(9 types), DataType(6 types)	7
FR2	Data Preprocessing and Transformation Pipeline The grammar should support: Sequential definition of data preprocessing steps including normalization , feature extraction , and data augmentation operations that can be chained together in a pipeline format.	PreprocessingSection, PreprocessingStep(11 variant rules → εδώ μέσα βρίσκεται το normalization, feature extraction, and data augmentation), CropStrategy(3 types), FlipDirection(3 types)	15
FR3	Training and Validation Data Split Configuration The grammar should support: Specification of data partitioning strategies for training, validation, and test sets with configurable split ratios and sampling methods .	PartitionSection, PartitionStrategy (4 types → εδώ καλύπτονται τα sampling methods)	2
FR4	Neural Network Architecture Definition The grammar should support: Hierarchical definition of network architectures through layer specification , including layer types , connections , and architectural patterns without exposing implementation details.	ArchitectureSection, ArchitectureType (3 types), ConnectionSpec, LayerSpec (14 variant rules)	18
FR5	Layer Parameter and Configuration Management The grammar should support: Declarative specification of layer parameters including	LayerSpecs (14 variant rules), WeightInitSpec (8 variant rules), RegularizerSpec (3 variant rules),	43

	dimensions, weights initialization , and layer-specific configurations in a human-readable format.	KernelSizeSpec(2 variant rules), StrideSpec(2 variant rules), PaddingType(3 types), PoolType(4 types), DropoutType(3 types), KeyValueParam, ParamValue(4 variant rules)	
FR6	Activation Function Selection The grammar should support: Selection and configuration of activation functions for different layers with parameterizable options for commonly used activation types .	ActivationSpec (12 variant rules)	13
FR7	Loss Function and Optimization Configuration The grammar should support: Specification of loss functions and optimization strategies with configurable parameters for learning behavior without requiring optimization theory knowledge.	LossSpec (9 variant rules), OptimizerSpec (8 variant rules), ReductionType (3 types)	20
FR8	Learning Rate and Training Schedule Management The grammar should support: Definition of learning rate policies, scheduling strategies , and decay parameters that can be easily configured for different training phases.	LRSchedulerSpec (11 variant rules), CyclicMode(3 types), AnnealStrategy(2 types), PlateauMode(2 types)	15
FR9	Batch Processing and Training Configuration The grammar should support: Specification of batch sizes, training epochs, and batch processing strategies with clear parameter definitions for training workflow control.	TrainingSection, BatchStrategy(4 types)	2
FR10	Model Training and Fitting Process Definition The grammar should support: Declarative specification of	TrainingSection, CheckpointSpec, CheckpointFrequency(4 types),	4

	the training process including training loops, checkpoint saving, and training monitoring without exposing training loop implementation.	CheckpointSaveMode(2 types)	
FR11	Model Evaluation and Metrics Configuration The grammar should support: Definition of evaluation metrics, scoring methods, and prediction output configurations for model assessment and validation.	EvaluationSection, MetricSpec (15 variant rules), AverageType(4 types), CurveType(2 types), PredictionOutputSpec	20
FR12	Transfer Learning and Pretrained Model Integration The grammar should support: Specification of pretrained model loading, layer freezing/unfreezing strategies, and model head replacement for transfer learning scenarios.	TransferLearningSection, BaseModelSpec, ModelSource(5 types), FreezeStrategySpec (3 variant rules), HeadReplacementSpec, FinetuningSpec	9
FR13	Convolutional and Recurrent Architecture Support The grammar should support: Specialized configuration for convolutional and recurrent network architectures including CNN layers, RNN variants, and LSTM configurations with domain-specific parameters.	ConvLayerSpec, PoolingLayerSpec, RecurrentLayerSpec, LSTMLayerSpec, GRULayerSpec(αυτά βρίσκονται μέσα στο LayerSpec) , PaddingType	6
FR14	Model Output and Prediction Configuration The grammar should support: Specification of model output formats, prediction post-processing, and result interpretation configurations for different use cases.	OutputSection, OutputFormat(6 types), PostProcessingStep (5 variant rules), InterpretationSpec (4 variant rules), ExportFormat (6 types)	14
FR15	Training Monitoring and Early Stopping Rules The grammar should support: Definition of training monitoring criteria, early	MonitoringSection, MonitoringCriterion (3 variant rules), EarlyStoppingRule, LoggingConfigSpec,	10

	stopping conditions, and validation-based training control rules that can trigger training modifications or termination.	LogFrequency(3 types), LoggerType(5 types), MonitorMode (2 types)	
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Total : 198 rules Average rules per FR : 13.2

Deep Learning – Standalone LLM

	FR context	Grammar Rules	Number of Grammar Rules
FR1	Dataset Definition and Semantics The grammar should support: Explicit declaration of datasets, including data sources, data splits (e.g., training, validation, evaluation), feature types , labels, and basic constraints (such as required fields or acceptable value ranges).	DatasetDecl, SplitDecl, SplitType(4 types), FeatureDecl, FeatureType(9 types), DatasetConstraint (3 variant rules)	9
FR2	Data Versioning Awareness The grammar should support: Referencing specific dataset versions or snapshots, enabling users to clearly associate training workflows with a well defined state of the data.	DatasetVersionDecl	1
FR3	Multi-Stage Training Workflow Modeling The grammar should support: The description of multi-stage training workflows, where each stage can depend on outputs or states produced by previous stages.	TrainingPipelineDecl, TrainingStageDecl	2
FR4	Stage-Level Configuration The grammar should support: Stage-specific configuration, allowing users to express that	TrainingStageDecl , ObjectiveDecl, DurationDecl	3

	different training stages may use different datasets, durations, objectives , or evaluation criteria.		
FR5	Model Structure Declaration The grammar should support: High-level model structure definitions , including the ability to declare layers , reusable blocks, and the ordering or connectivity between them.	ModelDecl, ModelBlockDecl, LayerDecl, LayerType (14 types)	4
FR6	Parameter Sharing Expression The grammar should support: Explicit expression of shared parameters, enabling users to indicate when multiple parts of a model reuse the same configurable components.	SharedParamDecl	1
FR7	Attention and Focus Mechanisms The grammar should support: The declaration of attention or focus mechanisms as named components that can be attached to relevant parts of the model structure.	AttentionDecl, AttentionType (7 types)	2
FR8	Modular Model Composition The grammar should support: Modular composition, allowing users to define, name, and reuse model substructures across different models or training workflows.	ModelBlockDecl	1
FR9	Pretrained State Initialization The grammar should support: Initializing a model or substructure from a predefined trained state, referenced symbolically rather than technically.	ModelDecl, FineTuningDecl, FineTuningStrategy(8 types)	3
FR10	Selective Freezing and Unfreezing The grammar should support: Declaring which model parts are fixed or adjustable during specific training stages ,	FreezeRuleDecl, FreezeAction (3 types) , TrainingStageDecl	3

	using intuitive, human-readable rules.		
FR11	Head Replacement and Extension The grammar should support: Replacing or extending the output portion of a model, enabling adaptation of an existing model structure to new prediction goals.	ModelHeadDecl, HeadType(6 types)	2
FR12	Retraining Trigger Definition The grammar should support: Declaration of retraining triggers, such as the arrival of new data , degradation of observed quality metrics, or detected changes in data characteristics .	RetrainingPolicyDecl, RetrainingTrigger(4 variant rules), DataUnit(6 types)	7
FR13	Metric-Based Conditions The grammar should support: Metric-based conditions, allowing users to express acceptable ranges, thresholds, or trends that influence retraining or workflow transitions.	MetricThresholdDecl, ComparisonOperator(6 types), TrendType(4 types), EvaluationDecl, MetricDecl	5
FR14	Data Change and Drift Awareness The grammar should support: Expressing sensitivity to data distribution changes, enabling users to specify when retraining should occur due to significant deviations from previous data behavior.	DataDriftTrigger, DriftDetectionMethod(6 types)	2
FR15	Validation of Conceptual Consistency The grammar should support: Validation rules that ensure conceptual consistency, such as ensuring referenced datasets exist, model parts are defined before use, and training stages form a coherent workflow.	Declaration	1

Total : 46 Average rules per FR : 3.06

ML over Data Streams - Methodology

	FR context	Grammar Rules	Number of Grammar Rules
FR1	Stream Source Configuration The grammar should support: Declaring multiple data stream sources with configurable connection parameters (endpoints, authentication, protocols), specifying source types (message queues, databases, file systems, APIs), and defining connection retry policies and error handling strategies.	StreamSource, StreamSourceType(10 types), StreamProtocol(10 types), AuthConfig(5 variant rules), RetryPolicy, ErrorHandlingStrategy, ErrorType(6 types), ConnectionParams, ErrorAction (5 types)	14
FR2	Schema Definition and Evolution The grammar should support: Defining structured schemas for incoming stream data with field names, types, and constraints, declaring schema evolution policies for handling new fields or type changes, and specifying backward compatibility rules when schemas change over time.	SchemaDefinition, SchemaField, FieldType(12 types), FieldConstraint(5 variant rules), SchemaEvolutionPolicy(8 types), CompatibilityRule (6 types)	11
FR3	Temporal Semantics Configuration The grammar should support: Specifying event-time versus processing-time semantics for stream events, defining timestamp extraction rules from incoming data records, and configuring watermark policies for handling late-arriving events in temporal processing.	TimestampSemantics(3 types), WatermarkPolicy, WatermarkStrategy(4 types), TimeUnit(5 types)	4
FR4	Window Type Specification The grammar should support: Declaring different	WindowSpec(4 variant rules), WindowTrigger(4 variant rules), CompositeTrigger,	12

	window types (tumbling, sliding, session-based) with configurable sizes and intervals, specifying window triggering conditions based on time, count, or custom predicates, and defining window overlap and alignment policies.	TriggerOperator	
FR5	Aggregation Operations Definition The grammar should support: Specifying statistical aggregations (sum, count, mean, quantiles, variance) over windowed data, defining custom aggregation functions for domain-specific metrics, and configuring incremental computation strategies for efficiency.	AggregationSpec, AggregationDef(2 variant rules), AggregationFunction(12 types), IncrementalStrategy(4 types)	6
FR6	Out-of-Order Event Handling The grammar should support: Defining policies for handling events that arrive outside their expected temporal order, specifying grace periods for accepting late events, and configuring strategies for updating previously computed results when late data arrives.	LateEventPolicy, LateEventAction(4 types), UpdateStrategy(4 types)	3
FR7	Feature Transformation Pipeline The grammar should support: Declaring feature extraction and transformation steps that operate on streaming data, specifying incremental feature computation methods that update with each new data point, and defining feature normalization and scaling	FeaturePipeline, FeatureStep(5 variant rules), ExtractionMethod(7 types), TransformOperation(11 types), NormalizationMethod(6 types), EncodingMethod(7 types), NormalizationConfig	12

	policies for streaming contexts.		
FR8	Online Model Training Configuration The grammar should support: Specifying online learning strategies that update models incrementally with streaming data, defining mini-batch sizes and update frequencies for model training , and configuring learning rate schedules and convergence criteria for streaming scenarios.	OnlineLearner, LearningAlgorithm(10 types), UpdateFrequency, UpdateFreqType(4 types), LearningRateSchedule, LRScheduleType(5 types), ConvergenceCriteria, TaskType(6 types)	8
FR9	Model State Management The grammar should support: Declaring how model parameters and state are maintained and updated during stream processing, specifying checkpointing policies for model persistence and recovery , and defining model versioning strategies for tracking changes over time .	ModelState, CheckpointPolicy, CheckpointFailureAction(4 types), VersioningStrategy, VersionScheme(4 types), StateBackend, StateBackendType(7 types), RetentionPolicy, CleanupStrategy(4 types)	9
FR10	Data Flow Orchestration The grammar should support: Defining end-to-end data processing pipelines that connect sources, transformations, models, and outputs , specifying data routing and branching logic for complex processing topologies, and declaring error handling and recovery mechanisms throughout the pipeline.	Pipeline, PipelineStage(8 variant rules), RoutingConfig, RoutingStrategy(5 types), RoutingRule, PipelineErrorHandling, SinkType (8 types)	15
FR11	Model Integration and Serving The grammar should support: Integrating trained models into streaming pipelines for real-time	InferenceSystem, FieldMapping, ServingPolicy, ServingStrategy(6 types), LoadBalancingStrategy(5 types), OutputFormat,	8

	inference , specifying model input/output mappings and data format conversions , and defining model serving policies including load balancing and scaling strategies .	DataFormat(7 types), CompressionType(5 types)	
FR12	Metrics and Monitoring Configuration The grammar should support: Declaring performance metrics to be computed and tracked during stream processing, specifying metric collection intervals and aggregation policies , and defining alerting thresholds and notification strategies for pipeline monitoring .	MonitoringConfig, MetricDefinition, MetricType(11 types), MetricAggregationPolicy, AlertRule, AlertCondition(7 types), AlertSeverity(4 types)	7
FR13	Real-time Prediction Serving The grammar should support: Configuring real-time prediction endpoints that serve model inferences on streaming data, specifying prediction output formats and delivery mechanisms , and defining latency requirements and performance guarantees for inference serving.	PredictionEndpoint, EndpointType(7 types), DeliveryMechanism, DeliveryType(5 types), LatencyRequirement, SLAPolicy(4 types), ScalingPolicy	7
FR14	Continuous Model Evaluation The grammar should support: Declaring continuous evaluation metrics that assess model performance on streaming data, specifying evaluation windows and comparison baselines for performance monitoring, and defining evaluation triggers based on data volume or time intervals.	EvaluationConfig, EvalMetricType(11 types), BaselineConfig, BaselineType(5 types), EvaluationTrigger, EvalTriggerType(4 types), ComparisonStrategy (4 types)	7
FR15	Drift Detection and Retraining Policies	DriftDetector, DriftDetectionMethod(9 types), DriftSensitivity(4	12

	The grammar should support: Configuring drift detection mechanisms that monitor changes in data distribution and model performance, specifying retraining triggers based on drift severity, performance degradation, or data volume thresholds, and defining retraining policies including data selection strategies and model update procedures.	types), RetrainingTrigger, RetrainingCondition(4 variant rules), RetrainingPolicy, DataSelectionStrategy(6 types), ModelUpdateProcedure(5 types)	
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Total : 135

Average rules per FR : 9

ML over Data Streams – Standalone LLM

	FR context	Grammar Rules	Number of Grammar Rules
FR1	Stream Source Definition The grammar should support: Declaring continuous data sources as streams, including their identifiers, data arrival mode (continuous or bursty), and connection parameters required to read incoming events.	StreamDeclaration, ArrivalMode(2 types), ConnectionParams(6 variant rules), ConnectionAuth (3 variant rules)	13
FR2	Stream Schema and Data Types The grammar should support: Defining the structure of streaming data, including fields, data types, optional attributes, and default values, with validation of schema consistency across pipeline stages.	SchemaDefinition, FieldDefinition, FieldType(2 variant rules), PrimitiveTypeName(10 types), DefaultValue(3 variant rules)	10
FR3	Time Semantics Specification	TimeSemanticDeclaration, TimeType(3 types),	7

	The grammar should support: Explicit declaration of time semantics for streams, including which field represents time , how time progresses, and whether processing is based on event time or processing time.	WatermarkStrategy(3 variant rules), DurationLiteral	
FR4	Schema and Stream Evolution Handling The grammar should support: Expressing how changes in stream structure or field availability are handled over time, including allowed additions, removals, or type changes.	SchemaEvolutionPolicy, CompatibilityMode(4 types), IncompatibleAction(4 types)	3
FR5	Window Definition and Configuration The grammar should support: Defining finite views over infinite streams using windows , including size, advance behavior, and alignment with the declared time semantics.	WindowDefinition, WindowType(4 types), WindowAlignment(3 types), DurationLiteral	4
FR6	Out-of-Order and Late Data Handling The grammar should support: Expressing acceptable delays in data arrival and how late or out-of-order events are treated within windowed or stateful processing.	WatermarkPolicy, LateEventAction(4 types)	2
FR7	Stateful Aggregation over Streams The grammar should support: Declaring stateful computations over streams, including grouped aggregations, maintained state scope, and rules for state update and expiration.	StatefulAggregation, AggregationExpression, AggFunction(10 types), StateScope(3 types), StateUpdateRule(4 types)	5
FR8	Incremental Feature Construction The grammar should support: Defining features that are continuously	FeatureDefinition, FeatureComputation, FeatureExpression(2 variant rules), TransformFunction(11	10

	updated as new events arrive , including transformations derived from raw stream fields and windowed or stateful values.	types), UpdateTrigger (3 variant rules)	
FR9	Online Model Definition and Binding The grammar should support: Declaring models that learn continuously from streams , including input features, target fields , and update frequency tied to incoming data.	OnlineModelDeclaration, MLAlgorithm(11 types), UpdateFrequency(3 variant rules), MLTaskType (5 types)	7
FR10	Pipeline Composition and Orchestration The grammar should support: Composing end-to-end pipelines by declaratively linking data streams, feature definitions, models, evaluation logic, and outputs into a single coherent flow.	PipelineDefinition	1
FR11	Real-Time Inference Configuration The grammar should support: Expressing how incoming stream events are scored in real time, including which model is used, when predictions are emitted , and where results are sent .	InferenceDeclaration, EmitCondition(3 variant rules), OutputSink(5 variant rules)	11
FR12	Continuous Evaluation Metrics The grammar should support: Defining evaluation metrics that are computed continuously over streaming predictions and outcomes, including their update scope and temporal aggregation .	MetricDefinition, MetricType(10 types), MetricUpdateScope(3 types), TemporalAggregation(3 types)	4
FR13	Data and Concept Drift Specification The grammar should support: Expressing conditions that indicate changes in data	DriftDetectorDeclaration, DriftType(5 types), DriftDetectionMethod(9 types), DriftSeverity(4 types), AlertCondition	5

	characteristics or prediction behavior over time, based on monitored stream statistics or evaluation signals.		
FR14	Retraining Triggers and Policies The grammar should support: Declaring retraining conditions driven by streaming criteria, such as accumulated data volume , sustained metric degradation, or persistent drift signals.	RetrainingPolicy, RetrainingTrigger(4 variant rules), RetrainingStrategy(4 types), VolumeUnit(5 types)	8
FR15	Validation and Consistency Constraints The grammar should support: Validating that declared streams, windows, features, models, metrics, and retraining rules are compatible in terms of schema, time semantics, and data dependencies.	No explicit validation rules	0

Total : 90

Average rules per FR : 6

NLP - Methodology

	FR context	Grammar Rules	Number of Grammar Rules
FR1	Text Data Source Definition and Ingestion The grammar should support: Declarative specification of text data sources including file paths, streaming inputs, database connections, and real-time feeds with configurable encoding formats and language detection.	DataSource(4 variant rules), FileFormat(10 types), StreamProtocol(6 types), DatabaseType(7 types), FeedType(5 types), AuthConfig, AuthType(5 types)	11

FR2	Tokenization Strategy Configuration The grammar should support: Specification of tokenization approaches including subword tokenization, sentence piece tokenization, word-level tokenization, and custom tokenization rules with vocabulary management.	Tokenizer (4 variant rules), SubwordAlgorithm(3 types), SentencePieceModelType(4 types), TokenizationRule	8
FR3	Text Preprocessing Pipeline Orchestration The grammar should support: Sequential definition of preprocessing steps including normalization, cleaning, stopword removal, stemming, lemmatization, and case handling with conditional execution paths.	Preprocessor, PreprocessingStep(7 variant rules), UnicodeForm(4 types), StemmingAlgorithm(4 types), LemmatizationBackend(5 types), CaseMode(4 types), ConditionalExecution, ErrorHandlingStrategy (4 types)	15
FR4	Feature Extraction and Representation The grammar should support: Configuration of text feature extraction methods including bag-of-words, TF-IDF, n-grams, and dense embeddings with dimensionality and sparsity controls.	FeatureExtractor (4 variant rules), NormType(3 types), AnalyzerType(3 types), PoolingStrategy(4 types), NGramRange	9
FR5	Embedding Model Integration and Management The grammar should support: Specification of embedding models including pre trained embeddings, contextual embeddings, positional embeddings, and custom embedding layers with fine-tuning capabilities.	EmbeddingModel (4 variant rules), PositionalEncodingType(4 types)	6

FR6	Named Entity Recognition Task Configuration The grammar should support: Definition of named entity recognition tasks including entity types, recognition boundaries , confidence thresholds, and custom entity categories with annotation schemas.	NERTask, EntityTypeDefinition, CustomEntityCategory, RecognitionBoundary(4 types)	4
FR7	Classification Task Specification The grammar should support: Configuration of text classification tasks including multi-class, multi-label, and hierarchical classification with class balancing and evaluation metrics.	ClassificationTask(3 variant rules), ClassBalancingStrategy(5 types), ClassificationMetric(8 types), ActivationFunction(5 types)	7
FR8	Model Training and Optimization Configuration The grammar should support: Specification of training procedures including batch sizes, learning rates, optimization strategies, regularization techniques, and convergence criteria .	TrainingConfig, OptimizerConfig, RegularizationConfig, ConvergenceCriteria, LRSchedulerConfig, OptimizerType(7 types), SchedulerType(7 types), ConvergenceMode (3 types)	8
FR9	Pipeline Workflow Orchestration The grammar should support: Definition of end-to-end pipeline workflows including component dependencies, data flow specifications, parallel processing, and error handling strategies .	Pipeline, PipelineComponent, DataFlowSpec, DataFlowMode(4 types), ParallelismConfig, ParallelStrategy(4 types), PipelineErrorHandling, PipelineDependency, PipelineComponentType(10 types)	9
FR10	Inference and Prediction Configuration The grammar should support: Specification of	InferenceMode(3 variant rules), DataFormat (8 types)	5

	inference modes including batch inference, real-time inference, streaming prediction , and confidence scoring with output formatting .		
FR11	Model Evaluation and Validation Setup The grammar should support: Definition of evaluation protocols including cross validation strategies, performance metrics, statistical significance testing, and benchmark comparisons .	EvaluationProtocol, ValidationStrategy(6 types), EvaluationMetric(14 types), SignificanceTestConfig, SignificanceMethod(5 types), CorrectionMethod(4 types), BenchmarkConfig	7
FR12	Language-Specific Processing Rules The grammar should support: Specification of language-specific processing including morphological analysis, language detection, script handling, and locale aware text processing rules .	LanguageProcessor, MorphologicalConfig, MorphologicalBackend(5 types), MorphologicalFeature(8 types), LanguageDetectionConfig, LanguageDetectionBackend(5 types), ScriptHandlingConfig, TextDirection(3 types), LocaleRule	9
FR13	Data Transformation and Augmentation The grammar should support: Configuration of data transformation operations including text augmentation, synthetic data generation, format conversion, and batch processing specifications .	DataTransformation, TransformationOperation (4 variant rules), AugmentationTechnique(7 types), SyntheticDataMethod(5 types)	8
FR14	Model Deployment and Serving Configuration The grammar should support: Specification of model deployment settings including serving endpoints, scaling policies, model versioning, checkpoint	DeploymentConfig, ServingEndpoint, ServingProtocol(5 types), ScalingPolicy, ScalingStrategy(4 types), ModelVersioningConfig, VersioningStrategy(4 types), CheckpointConfig, CheckpointFormat(5 types),	13

	management, and performance monitoring.	MonitoringConfig, MonitoringMetric(7 types), AlertThreshold, DeploymentEnvironment (4 types)	
FR15	Annotation and Labeling Schema Definition The grammar should support: Definition of annotation schemas including label taxonomies, annotation guidelines, inter-annotator agreement metrics, and quality assurance protocols for training data preparation.	AnnotationSchema, TaxonomyNode, QualityProtocol, AgreementMetric(5 types), LabelFormat(8 types), ReviewProcess(4 types), AdjudicationStrategy(4 types)	7

Total : 126

Average rules per FR : 8.4

NLP – Standalone LLM

	FR context	Grammar Rules	Number of Grammar Rules
FR1	Text Data Definition The grammar should support: Defining text data sources, text fields, languages, encodings, and document boundaries as first-class concepts.	DataSource, DataSourceType(8 types), DataFormat(10 types), EncodingType(6 types), TextField, DocumentBoundary(8 types)	6
FR2	Text Segmentation and Normalization The grammar should support: Declarative specification of text segmentation, normalization rules, and cleanup steps applied before analysis.	Preprocessor, PreprocessingStep (4 variant rules), NormalizationType(10 types), CleanupType(9 types), SegmentationType(7 types), SegmentationConfig, NormParam	11
FR3	Linguistic Unit Configuration The grammar should support: Expressing which	LinguisticUnit(7 types)	1

	linguistic units (such as tokens, sentences, phrases, or documents) are used at each stage of processing.		
FR4	Language Awareness and Multilingual Handling The grammar should support: Explicit declaration of language-specific behaviors and multilingual processing paths within a single pipeline.	LanguageConfig, TokenizerType(7 types), SentenceDetectorType(4 types), StopwordConfig(3 types), TextDirection(3 types), MultilingualPath	6
FR5	Linguistic Representation Selection The grammar should support: Specifying how text is represented linguistically (e.g., symbolic, contextual, or semantic representations) without exposing technical complexity.	Representation, RepresentationType(9 types), GranularityLevel(6 types)	3
FR6	Embedding Abstraction The grammar should support: Configuring text embedding choices as abstract, human-readable options suitable for different tasks and languages.	EmbeddingConfig, EmbeddingModel(10 types), EmbeddingTask(7 types), PoolingStrategy(5 types)	4
FR7	Task-Centric Workflow Definition The grammar should support: Defining end-to-end NLP workflows centered on user intended tasks rather than technical processing steps.	Workflow, WorkflowTask(12 types), ProcessingMode(5 types)	3
FR8	Information Extraction Task Specification The grammar should support: Declarative definition of information extraction tasks, including entity detection, entity classification, and relationship identification.	ExtractionTask, ExtractionType(8 types), EntityTypeSpec, RelationSpec	4

FR9	Annotation and Label Semantics The grammar should support: Defining label sets , annotation meanings, constraints , and relationships in a way that is understandable to non-experts.	LabelSet, LabelDefinition, LabelTask(10 types), LabelConstraint, LabelConstraintType(7 types), LabelRelationship, RelationshipType(6 types)	7
FR10	Training and Inference Mode Declaration The grammar should support: Clear differentiation between training, validation, and inference configurations over text pipelines.	TrainingConfig, TrainingPhase(6 types), AdaptationMode(6 types), CheckpointStrategy(4 types)	4
FR11	Pipeline Composition and Orchestration The grammar should support: Composing, ordering, reusing, and branching NLP pipeline stages as a single coherent specification.	Pipeline, PipelineStage(2 variant rules), StageType(11 types), PipelineBranch	6
FR12	Real-Time and Batch Text Processing The grammar should support: Expressing whether text processing and inference occur in real-time, batch, or hybrid modes.	ProcessingMode(5 types)	1
FR13	Quality and Performance Signals The grammar should support: Defining observable quality signals and acceptance criteria related to text understanding and task outcomes.	QualitySignal, QualityMetric(14 types), QualityComparator(5 types), AcceptanceCriterion	4
FR14	Language and Topic Drift Awareness The grammar should support: Expressing conditions under which changes in language usage, vocabulary, or topics are	DriftPolicy, DriftDetectionMethod(7 types), LinguisticAspect(8 types), DriftSensitivity(4 types), DriftAction(6 types)	5

	detected and acknowledged.		
FR15	Retraining and Adaptation Triggers The grammar should support: Declarative rules that describe when and why retraining or adaptation is required based on data changes, drift, or quality signals.	RetrainingPolicy, RetrainingTrigger, RetrainingTriggerType(8 types), RetrainingSchedule(6 types), RetrainingCondition, RetrainingConditionType(6 types), RetrainingPriority(4 types)	7

Total : 72

Average rules per FR : 4.8

Total Analysis

FRs Set	Total rules	Avg rules per FR
DL MCP	198	13.2
DL LLM	46	3.06
ML Streams MCP	135	9
ML Streams LLM	90	6
NLP MCP	126	8.4
NLP LLM	72	4.8